



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Computer Vision

Course Code: ITA0506

Slot: D

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1. Gradient Magnitude and Direction

The gradient magnitude is calculated using:

The gradient direction is:

For Q1:

Given:

$$G_x = 24$$

$$G_y = 18$$

Gradient Magnitude

$$G = \sqrt{24^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30$$

Gradient Direction

$$\theta = \tan^{-1}\left(\frac{24}{18}\right) = \tan^{-1}(1.33) \approx 53.13^\circ$$

So

$$\text{Gradient Magnitude} = 30$$

$$\text{Gradient Direction} = 53.13^\circ$$

2. Non-Maximum Suppression

Given:

Pixel	P1	P2	P3	P4	P5
Magnitude	25	50	90	60	30

Gradient direction is horizontal, so each pixel is compared with its left and right neighbors.

P1 = 25 → boundary

P2 = 50 → less than P3 (90) → suppress

P3 = 90 → greater than P2 (50) and P4 (60) → keep

P4 = 60 → less than P3 (90) → suppress

P5 = 30 → boundary

Result:

Pixel	P1	P2	P3	P4	P5
After NMS	0	0	90	0	0

Answer: P3 is the only retained edge pixel.

3. Double Thresholding

Given:

High Threshold = 100

Low Threshold = 50

Pixel	Magnitude	Classification
P1	135	Strong Edge
P2	90	Weak Edge
P3	45	Non-edge

P4	120	Strong Edge
P5	70	Weak Edge
P6	20	Non-edge

Rules

Magnitude $\geq 100 \rightarrow$ Strong Edge

$50 \leq \text{Magnitude} < 100 \rightarrow$ Weak Edge

Magnitude $< 50 \rightarrow$ Non-edge

Final Answer

Strong edges: P1, P4

Weak edges: P2, P5

Non-edges: P3, P6

4. Hysteresis Thresholding

Given:

High Threshold = 100

Low Threshold = 50

Pixel	Magnitude	Classification
P1	130	Strong
P2	95	Weak
P3	40	Non-edge
P4	115	Strong
P5	65	Weak
P6	25	Non-edge
P7	105	Strong

1. Strong Edges

Magnitude ≥ 100 :

P1, P4, P7

2. Weak Edges

$50 \leq \text{magnitude} < 100$:

P2, P5

3. Non-edges

Magnitude < 50 :

P3, P6

4. Final Edge Pixels After Hysteresis

Given:

P2 is connected to P1 $\rightarrow$ P2 is connected to a strong edge, so keep P2.

P5 is not connected to any strong edge $\rightarrow$ remove P5.

Therefore:

Final edge pixels = P1, P2, P4, P7

Final Answer

Category	Pixels
Strong edges	P1, P4, P7
Weak edges	P2, P5
Non-edges	P3, P6
Final edges	P1, P2, P4, P7